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Professor Minsoo Kang, Editor-in-Chief
Measurement in Physical Education and Exercise Science

Dear Professor Kang,

Submission of an Original Article: "How much is the wind worth? Sprints and horizontal jumps from age 10 to senior level"

We submit the manuscript above for consideration as an Original Article in *Measurement in Physical Education and Exercise Science*, to be reviewed in the Exercise Science section.

A sprint or jump mark is meant to measure the athlete, but it also measures the wind. Athletics regulates this source of error with a single ± 2.0 m/s legality rule that applies to ten-year-olds as to Olympic finalists, yet almost all empirical estimates of the wind effect come from elite senior athletes. Using every eligible wind-measured sprint and horizontal-jump result in the Norwegian Athletics Federation's registry from age 10 to senior level, the manuscript estimates the wind effect within athlete and season and shows how it scales with age, ability and event. Per meter per second the effect on sprint times is clearly larger for children than for seniors, mainly because slower athletes are exposed to the wind for longer; headwinds cost more than tailwinds give, by about as much as drag physics predicts on the 100 m; and a legal tailwind is worth more than a tenth of a second on the 100 and 200 m in every age band. The same estimator applied to US collegiate and championship-finalist samples places the elite literature at the fast end of the gradient. The results supply age- and event-specific wind corrections for youth lists, records and talent evaluation, and we believe they will interest the journal's readers working on performance measurement.

The analysis plan was committed to version control before estimation (the study was not externally preregistered), and departures from it are listed in the supplementary material. All estimates were reproduced by an independent implementation in a second programming language. The study was conducted in Norway, where it is not health research under the Health Research Act and required no ethics committee approval; Sikt (the Norwegian Agency for Shared Services in Education and Research) confirmed that the use of the data raises no data-protection concerns. The Methods state this in full, as the Taylor & Francis ethics policy provides for studies that do not require approval. The analysis code will be deposited in a public repository on acceptance.

In the journal's format (double-spaced, each element on a new page) the manuscript is 30 pages, with about 5,500 words of main text, four tables and three figures. The supplementary material, in a separate file, holds supplementary methods, ten supplementary tables and two supplementary figures.

The manuscript has not been published elsewhere and has not been submitted simultaneously for publication elsewhere, and both authors have approved the submission. The authors report no competing interests and received no grants or other funding for this work. The use of generative AI is declared in the manuscript. The registry also underlies other work by the corresponding author on youth performance development and participation (one published study is cited in the manuscript); none of that work examines wind.

Yours sincerely,

Atle G. Guttormsen (corresponding author), on behalf of both authors